04.17 ComGS复现

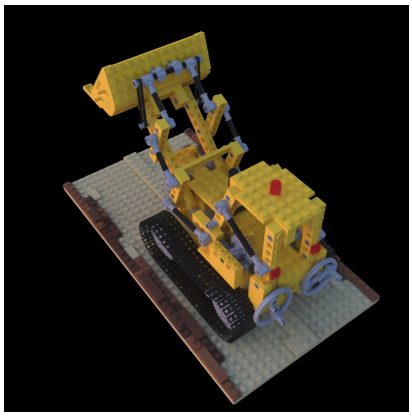
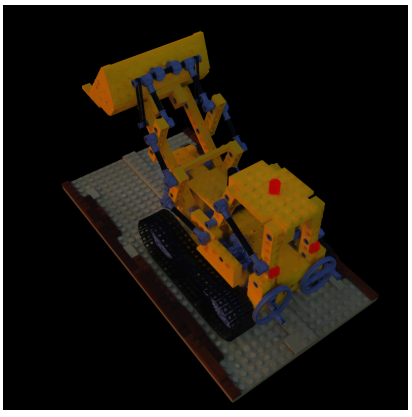
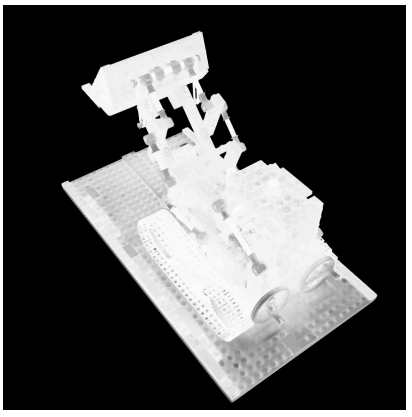
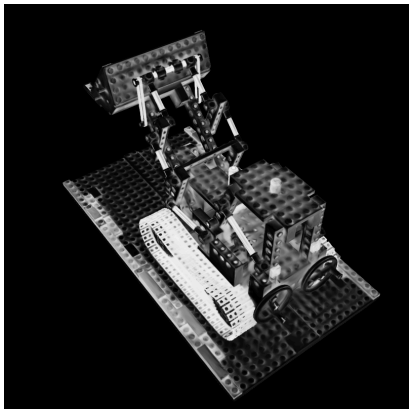
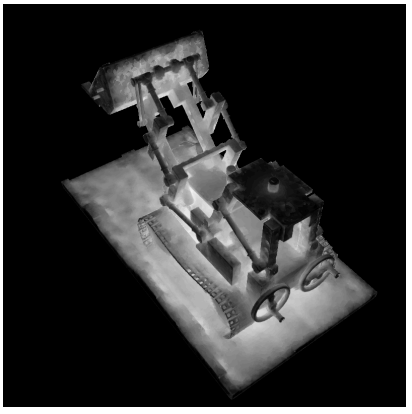
流程

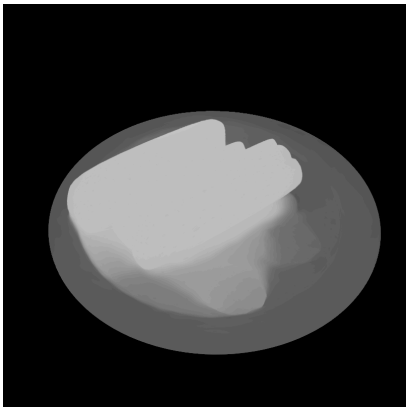
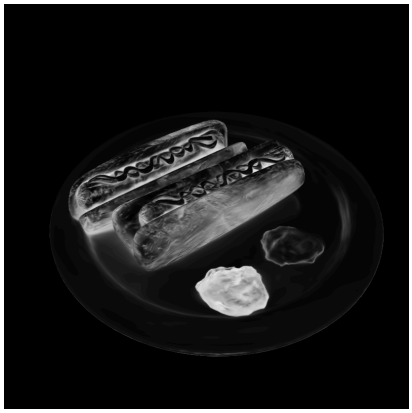
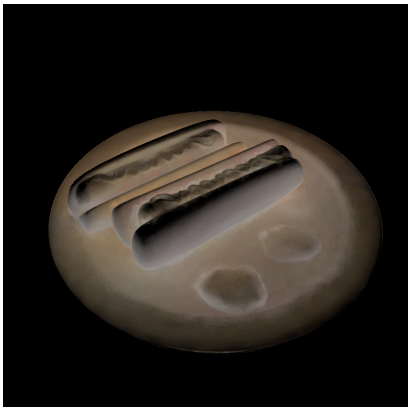
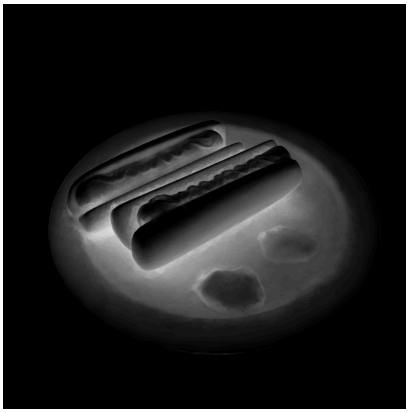
- 一阶段还是参考IRGS的方法，用RefGS把几何训练完成
- 二阶段初始化了SOP之后，冻结几何和SH系数，采用全像素计算PBR的方法计算（follow 论文里面的实现方法），计算2000轮。

定量与定性结果

目前复现的结果可以实现相对可行的效果，以下为TensoIR数据集上的结果（A800上测试）。

IRGS				COMGS		
	PSNR	SSIM	LPIPS	PSNR	SSIM	LPIPS
armadillo	39.86	0.977	0.047	39.20	0.978	0.045
ficus	35.74	0.981	0.025	36.05	0.983	0.022
hotdog	35.18	0.966	0.052	35.41	0.971	0.044
lego	31.04	0.930	0.074	32.09	0.945	0.055
average	35.45	0.964	0.050	35.69	0.969	0.042

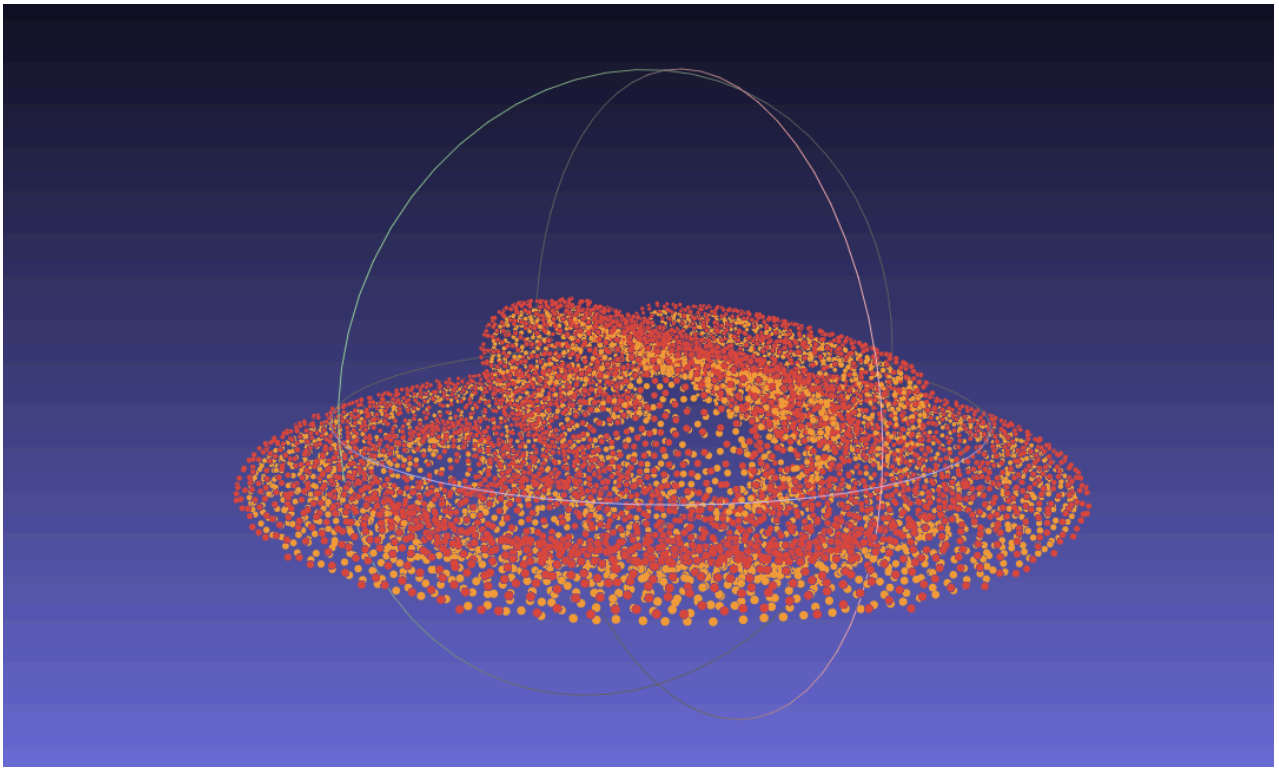
RENDER/METALLIC	ALBEDO/INDIRECT	ROUGNNESS/OCCU
		
		

RENDER/METALLIC	ALBEDO/INDIRECT	ROUGNNESS/OCCU
		
		

工程细节

复现过程中主要困难的地方在于SOP探针的初始化和超参数的设定上。

- SOP探针初始化：以hotdog为例，黄色的点为深度反投影得到的，红色是加入偏移之后得到的探针，如果不注意法线的方向，盘子周围的探针会移动到表面点的底部，导致光照计算不准。



- 超参数的设定：主要还是基本沿用IRGS的超参数，逆渲染框架下面修改太大容易掉点明显，调参数花费了一些时间。主要就是一些诸如roughness，metallic的初始化参数，各种学习率。

技术细节

- 效率分析：目前复现的方法没办法保证实时性。虽然说用上SOP之后，已经能够每轮是全像素计算PBR loss，但效率还是没办法达到论文里面宣称，包括训练时间在8min内，relight只需要7ms完成。现在不考虑relight，单纯考虑NVS任务，render一帧的时长仍然需要600ms左右。主要的问题在于：
- knn需要消耗一定时间(100 ms左右)，因为每一次 shading point 从 depth 反投影回去，都需要用 knn 进行查询邻近的 probes，这是非常耗时的。像 DDGI 这种方案，基本都是固定的 volume 顶点放置 probes，这样每次shading point都是$O(1)$的常数查询时间，然后三线性插值即可。
- SOP query的时间(250 ms左右)：每次N个shading point发射P条光线，需要从邻近k个SOP中查询间接光照和遮挡，需要$O(N \times P \times k)$个查询，总体也比较耗时。
- 最后环境光查询+蒙特卡洛估计整合得到像素rgb（200 ms左右）。
- 现在render还是基于Hammersley采样，没有上论文里面的多重重要性采样，之前尝试过，但噪声问题导致是需要去噪的，这里的细节文章也没有详细说明，但在IRGS++(ICLR2026被拒了，IRGS原作者的工程增量式工作)里面，有介绍一些细节。后续可以继续参考。

3.3.1 Multiple Importance Sampling

Multiple importance sampling Veach & Guibas (1995) (MIS) provides a methodology for combining samples from multiple probability distributions, enabling the sampling distribution to approximate the characteristics of target integrand. The estimator with balance heuristic is formulated as:

$$\sum_{i=1}^{n} \frac{1}{n_i} \sum_{j=1}^{n_i} m_i(X_{i,j}) \frac{g(X_{i,j})}{p_i(X_{i,j})}, \qquad m_i(x) = \frac{n_i p_i(x)}{\sum_k n_k p_k(x)}. \tag{6}$$

To reduce variance when evaluating the rendering equation, we implement MIS with three distinct sampling strategies: 1) Cosine-weighted distribution targeting the diffuse component, 2) GGX distribution Heitz (2018) aligned with the specular lobe, and 3) Environment light distribution generated through intensity-based sampling of environment map Pharr & Humphreys (2010). The GGX sampling proves particularly effective for glossy surfaces by concentrating samples around specular reflection directions, enabling accurate estimation of specular contributions in $\boldsymbol{c}_{\text{pbr}}$ with only a few samples. Light sampling handles strong directional illumination, significantly eliminates artifacts.

3.3.2 Denoising

Denoising in computer graphics enables high-quality rendering with low sample counts by reducing noise in Monte Carlo estimators, thereby enhancing the stability and efficiency of the rendering process. Typical denoising implementations employ spatial filter kernels that perform low-pass operations on noisy inputs through weighted neighborhood averaging. However, while aiming to use as few samples as possible in Monte Carlo integration, the limited sampling rate inevitably results in images with substantial noise. Inspired by Nvdiffrec-MC Hasselgren et al. (2022), we leverage the image denoising technique, using a cross-bilateral filter based on Spatio-temporal Variance-Guided Filtering Schied et al. (2017) (SVGF), which preserves geometric edges through depth and normal-aware weighting. The bilateral weighting between pixels p and q is computed as:

$$\text{Bilateral}\,(p,q) = e^{-\frac{|p-q|^2}{2\sigma^2}} e^{-\frac{|z(p)-z(q)|}{\sigma_z |\nabla z(p)\cdot(p-q)|}} \max(0, \boldsymbol{n}(p)\cdot\boldsymbol{n}(q))^{\sigma_n}, \tag{7}$$

where z, $\boldsymbol{n}$ denote the image space depth and surface normal, respectively.

总结

目前的工作后续relight也只能保证物体静止的时候渲染的实时性，当在场景中移动物体的时候，这里的实时性很难保证，如果想沿着 probe 的思路继续下去，应该需要更多参考DDGI的方法。